

Pandas Analysis Project: Superstore Sales Performance Analysis

1. Recommended Dataset

Primary Dataset

Dataset name: Superstore Sales Dataset

Source: Kaggle

Dataset link: <https://www.kaggle.com/datasets/rohitsahoo/sales-forecasting>

Why This Dataset Is Good

This dataset is very suitable for a beginner-to-intermediate **data analyst portfolio project** because it contains realistic retail business data. You can use it to practice:

- Data cleaning
- Exploratory data analysis
- Sales analysis
- Profit analysis
- Product analysis
- Customer analysis
- Regional analysis
- Discount impact analysis
- Time-series trend analysis
- Business insight writing

Dataset Language

The dataset is in **English**.

Column names and values such as product category, region, segment, ship mode, sales, profit, and order date are in English.

2. Alternative English Datasets

You can also use these datasets:

Option 1: Superstore Dataset - Sales & Profits of a Superstore

Link: <https://www.kaggle.com/datasets/vivek468/superstore-dataset-final>

Good for:

- Sales analysis
- Profit analysis
- Category analysis
- Region analysis
- Customer segment analysis

Option 2: Online Sales Dataset - Popular Marketplace Data

Link: <https://www.kaggle.com/datasets/shreyanshverma27/online-sales-dataset-popular-marketplace-data>

Good for:

- E-commerce analysis
 - Product category analysis
 - Payment method analysis
 - Regional analysis
 - Revenue analysis
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3. Project Title

Superstore Sales and Profitability Analysis Using Python Pandas

4. Project Goal

The goal of this project is to analyze retail sales data and discover business insights related to:

- Sales performance

- Profitability
- Product categories
- Customer segments
- Regional performance
- Monthly and yearly trends
- Shipping behavior
- Discount impact
- Loss-making products or regions

This project should show that you can clean data, analyze business questions, create useful visualizations, and explain insights like a real data analyst.

5. Tools Required

Use the following tools:

- Python
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Jupyter Notebook
 - Optional: Plotly
-

6. Suggested Project Folder Structure

```
superstore-sales-analysis/  
|  
├── data/  
│   └── superstore.csv  
|  
├── notebooks/  
│   └── superstore_analysis.ipynb  
|  
├── images/  
│   └── saved_plots/  
|  
└── README.md
```

7. Expected Dataset Columns

Depending on the Kaggle version, the dataset may include columns like:

- Order ID
 - Order Date
 - Ship Date
 - Ship Mode
 - Customer ID
 - Customer Name
 - Segment
 - Country
 - City
 - State
 - Region
 - Product ID
 - Category
 - Sub-Category
 - Product Name
 - Sales
 - Quantity
 - Discount
 - Profit
-

8. Main Business Questions

A. Basic Data Understanding

1. How many rows and columns are in the dataset?
2. What are the data types of each column?
3. Are there any missing values?
4. Are there duplicate records?
5. What is the time range of the dataset?

6. How many unique customers are there?
 7. How many unique products are there?
 8. How many product categories and sub-categories are present?
-

B. Sales Performance Questions

1. What is the total sales amount?
 2. What is the total profit?
 3. What is the total quantity sold?
 4. What is the average sales amount per order?
 5. Which year had the highest sales?
 6. Which month had the highest sales?
 7. What is the monthly sales trend?
 8. Are sales increasing or decreasing over time?
 9. Which quarter performs best in terms of sales?
 10. Which day of the week has the highest sales?
-

C. Profit Analysis Questions

1. What is the overall profit margin?
 2. Which year generated the highest profit?
 3. Which month generated the highest profit?
 4. Which categories are most profitable?
 5. Which sub-categories are least profitable?
 6. Which products generated the highest profit?
 7. Which products caused the highest losses?
 8. Are high sales always linked with high profit?
 9. Which states or regions are losing money?
 10. What is the relationship between discount and profit?
-

D. Product Analysis Questions

1. Which category has the highest sales?
2. Which category has the highest profit?
3. Which sub-category has the highest sales?
4. Which sub-category has the highest profit?
5. Which products are sold most frequently?

6. Which products generate the most revenue?
 7. Which products generate the most profit?
 8. Which products have high sales but low profit?
 9. Which products should the business focus on?
 10. Which products may need pricing or discount changes?
-

E. Customer Analysis Questions

1. Which customer segment generates the highest sales?
 2. Which customer segment generates the highest profit?
 3. Who are the top 10 customers by sales?
 4. Who are the top 10 customers by profit?
 5. Are some customers generating losses?
 6. What is the average order value by customer segment?
 7. Which segment buys the most quantity?
 8. Which segment receives the highest discount?
 9. Which customers are repeat buyers?
 10. Which customer segment should the company target more?
-

F. Regional Analysis Questions

1. Which region has the highest sales?
 2. Which region has the highest profit?
 3. Which region has the lowest profit?
 4. Which state has the highest sales?
 5. Which state has the highest loss?
 6. Which city has the highest sales?
 7. Which cities are most profitable?
 8. Which regions have high sales but low profit?
 9. Which region gives the highest average discount?
 10. Which region should the business improve?
-

G. Shipping Analysis Questions

1. Which shipping mode is used the most?
2. Which shipping mode generates the highest sales?
3. Which shipping mode generates the highest profit?

4. What is the average shipping time?
 5. Does shipping time affect profit?
 6. Which shipping mode is fastest?
 7. Which shipping mode is slowest?
 8. Which customer segment uses which shipping mode most?
 9. Are urgent shipping modes more profitable?
 10. Which shipping mode should the company promote?
-

H. Discount Analysis Questions

1. What is the average discount?
 2. Which category receives the highest discount?
 3. Which region receives the highest discount?
 4. Does discount increase sales?
 5. Does discount reduce profit?
 6. What discount range causes loss?
 7. Which products are heavily discounted?
 8. Which sub-categories are unprofitable due to discounts?
 9. Should the company reduce discounts in some areas?
 10. What is the best discount range for profit?
-

9. Data Cleaning Tasks

Before answering the questions, perform these cleaning steps:

1. Load the dataset using pandas.
2. Check column names.
3. Rename columns if needed.
4. Convert `Order Date` and `Ship Date` to datetime format.
5. Remove duplicate rows.
6. Check missing values.
7. Handle missing values if present.
8. Create new columns:
 - `Year`
 - `Month`
 - `Month Name`
 - `Quarter`
 - `Day of Week`

- Shipping Days
- Profit Margin

9. Check for negative profit values.

10. Check for unusual discounts or sales values.

10. Feature Engineering Ideas

Create useful columns for analysis:

```
df["Order Date"] = pd.to_datetime(df["Order Date"])
df["Ship Date"] = pd.to_datetime(df["Ship Date"])

df["Year"] = df["Order Date"].dt.year
df["Month"] = df["Order Date"].dt.month
df["Month Name"] = df["Order Date"].dt.month_name()
df["Quarter"] = df["Order Date"].dt.quarter
df["Day of Week"] = df["Order Date"].dt.day_name()

df["Shipping Days"] = (df["Ship Date"] - df["Order Date"]).dt.days
df["Profit Margin"] = df["Profit"] / df["Sales"]
```

11. Important KPIs to Calculate

KPI	Formula
Total Sales	Sum of Sales
Total Profit	Sum of Profit
Total Orders	Count of unique Order IDs
Total Customers	Count of unique Customer IDs
Total Quantity Sold	Sum of Quantity
Average Order Value	Total Sales / Total Orders
Profit Margin	Total Profit / Total Sales
Average Discount	Mean of Discount
Average Shipping Days	Mean of Shipping Days

Example code:

```
total_sales = df["Sales"].sum()
total_profit = df["Profit"].sum()
total_orders = df["Order ID"].nunique()
total_customers = df["Customer ID"].nunique()
total_quantity = df["Quantity"].sum()
average_order_value = total_sales / total_orders
profit_margin = total_profit / total_sales
average_discount = df["Discount"].mean()
average_shipping_days = df["Shipping Days"].mean()

kpi_summary = {
    "Total Sales": total_sales,
    "Total Profit": total_profit,
    "Total Orders": total_orders,
    "Total Customers": total_customers,
    "Total Quantity Sold": total_quantity,
    "Average Order Value": average_order_value,
    "Profit Margin": profit_margin,
    "Average Discount": average_discount,
    "Average Shipping Days": average_shipping_days
}

kpi_summary
```

12. Plot Ideas to Build

Plot 1: Monthly Sales Trend

Chart type: Line chart

Question answered: How do sales change over time?

```
monthly_sales = df.groupby(df["Order Date"].dt.to_period("M"))["Sales"].sum()

monthly_sales.plot(
    kind="line",
    figsize=(12, 5),
    title="Monthly Sales Trend"
)
```

Plot 2: Yearly Sales and Profit

Chart type: Bar chart

Question answered: Which year performed best?

```
yearly_performance = df.groupby("Year")["Sales", "Profit"].sum()

yearly_performance.plot(
    kind="bar",
    figsize=(10, 5),
    title="Yearly Sales and Profit"
)
```

Plot 3: Sales by Category

Chart type: Bar chart

Question answered: Which category has the highest sales?

```
category_sales = df.groupby("Category")["Sales"].sum().sort_values(ascending=True)

category_sales.plot(
    kind="bar",
    figsize=(8, 5),
    title="Sales by Category"
)
```

Plot 4: Profit by Category

Chart type: Bar chart

Question answered: Which category is most profitable?

```
category_profit = df.groupby("Category")["Profit"].sum().sort_values(ascending=True)

category_profit.plot(
    kind="bar",
    figsize=(8, 5),
)
```

```
title="Profit by Category"
```

```
)
```

Plot 5: Sales by Sub-Category

Chart type: Horizontal bar chart

Question answered: Which sub-categories generate the most sales?

```
subcategory_sales = df.groupby("Sub-Category")["Sales"].sum().sort_values
```

```
subcategory_sales.plot(  
    kind="barh",  
    figsize=(10, 7),  
    title="Sales by Sub-Category"  
)
```

Plot 6: Profit by Sub-Category

Chart type: Horizontal bar chart

Question answered: Which sub-categories are profitable or unprofitable?

```
subcategory_profit = df.groupby("Sub-Category")["Profit"].sum().sort_valu
```

```
subcategory_profit.plot(  
    kind="barh",  
    figsize=(10, 7),  
    title="Profit by Sub-Category"  
)
```

Plot 7: Sales by Region

Chart type: Bar chart

Question answered: Which region has the highest sales?

```
region_sales = df.groupby("Region")["Sales"].sum().sort_values(ascending=
```

```
region_sales.plot(  

```

```
kind="bar",  
figsize=(8, 5),  
title="Sales by Region"  
)
```

Plot 8: Profit by Region

Chart type: Bar chart

Question answered: Which region has the highest profit?

```
region_profit = df.groupby("Region")["Profit"].sum().sort_values(ascending=  
  
region_profit.plot(  
    kind="bar",  
    figsize=(8, 5),  
    title="Profit by Region"  
)
```

Plot 9: Sales by Customer Segment

Chart type: Bar chart

Question answered: Which customer segment contributes most to sales?

```
segment_sales = df.groupby("Segment")["Sales"].sum().sort_values(ascending=  
  
segment_sales.plot(  
    kind="bar",  
    figsize=(8, 5),  
    title="Sales by Customer Segment"  
)
```

Plot 10: Profit by Customer Segment

Chart type: Bar chart

Question answered: Which customer segment is most profitable?

```
segment_profit = df.groupby("Segment")["Profit"].sum().sort_values(ascending=True)

segment_profit.plot(
    kind="bar",
    figsize=(8, 5),
    title="Profit by Customer Segment"
)
```

Plot 11: Discount vs Profit

Chart type: Scatter plot

Question answered: Does discount reduce profit?

```
df.plot(
    kind="scatter",
    x="Discount",
    y="Profit",
    figsize=(8, 5),
    title="Discount vs Profit"
)
```

Plot 12: Sales vs Profit

Chart type: Scatter plot

Question answered: Do higher sales always mean higher profit?

```
df.plot(
    kind="scatter",
    x="Sales",
    y="Profit",
    figsize=(8, 5),
    title="Sales vs Profit"
)
```

Plot 13: Top 10 Products by Sales

Chart type: Horizontal bar chart

Question answered: Which products generate the highest revenue?

```
top_products_sales = (  
    df.groupby("Product Name")["Sales"]  
    .sum()  
    .sort_values(ascending=False)  
    .head(10)  
)  
  
top_products_sales.sort_values().plot(  
    kind="barh",  
    figsize=(10, 6),  
    title="Top 10 Products by Sales"  
)
```

Plot 14: Top 10 Products by Profit

Chart type: Horizontal bar chart

Question answered: Which products generate the highest profit?

```
top_products_profit = (  
    df.groupby("Product Name")["Profit"]  
    .sum()  
    .sort_values(ascending=False)  
    .head(10)  
)  
  
top_products_profit.sort_values().plot(  
    kind="barh",  
    figsize=(10, 6),  
    title="Top 10 Products by Profit"  
)
```

Plot 15: Top 10 Loss-Making Products

Chart type: Horizontal bar chart

Question answered: Which products are causing the most loss?

```
loss_products = (
    df.groupby("Product Name") ["Profit"]
      .sum()
      .sort_values()
      .head(10)
)

loss_products.sort_values().plot(
    kind="barh",
    figsize=(10, 6),
    title="Top 10 Loss-Making Products"
)
```

Plot 16: Average Discount by Category

Chart type: Bar chart

Question answered: Which category receives the highest discount?

```
discount_category = df.groupby("Category") ["Discount"].mean().sort_values()

discount_category.plot(
    kind="bar",
    figsize=(8, 5),
    title="Average Discount by Category"
)
```

Plot 17: Shipping Mode Count

Chart type: Bar chart

Question answered: Which shipping mode is most used?

```
ship_mode_count = df["Ship Mode"].value_counts()

ship_mode_count.plot(
    kind="bar",
    figsize=(8, 5),
    title="Shipping Mode Usage"
)
```

Plot 18: Average Shipping Days by Ship Mode

Chart type: Bar chart

Question answered: Which shipping mode is fastest?

```
shipping_days = df.groupby("Ship Mode")["Shipping Days"].mean().sort_valu

shipping_days.plot(
    kind="bar",
    figsize=(8, 5),
    title="Average Shipping Days by Ship Mode"
)
```

Plot 19: Sales Heatmap by Region and Category

Chart type: Heatmap

Question answered: Which region-category combination performs best?

```
import seaborn as sns
import matplotlib.pyplot as plt

pivot_sales = df.pivot_table(
    values="Sales",
    index="Region",
    columns="Category",
    aggfunc="sum"
)

plt.figure(figsize=(8, 5))
sns.heatmap(pivot_sales, annot=True, fmt=".0f", cmap="Blues")
plt.title("Sales Heatmap by Region and Category")
plt.show()
```

Plot 20: Profit Heatmap by Region and Category

Chart type: Heatmap

Question answered: Which region-category combination is most profitable?


```
pivot_profit = df.pivot_table(  
    values="Profit",  
    index="Region",  
    columns="Category",  
    aggfunc="sum"  
)  
  
plt.figure(figsize=(8, 5))  
sns.heatmap(pivot_profit, annot=True, fmt=".0f", cmap="Greens")  
plt.title("Profit Heatmap by Region and Category")  
plt.show()
```

13. Advanced Analysis Ideas

A. Customer Segmentation

Group customers based on:

- Total sales
- Total profit
- Number of orders
- Average order value

Possible customer types:

- High-value customers
- Frequent buyers
- Low-profit customers
- Loss-making customers

Example:

```
customer_summary = df.groupby("Customer ID").agg({  
    "Sales": "sum",  
    "Profit": "sum",  
    "Order ID": "nunique",  
    "Quantity": "sum"  
}).rename(columns={"Order ID": "Total Orders"})  
  
customer_summary["Average Order Value"] = (
```

```
customer_summary["Sales"] / customer_summary["Total Orders"]
)

customer_summary.head()
```

B. Product Performance Segmentation

Group products into:

- High sales, high profit
- High sales, low profit
- Low sales, high profit
- Low sales, low profit

This is useful because not every high-sales product is good for business.

Example idea:

```
product_summary = df.groupby("Product Name").agg({
    "Sales": "sum",
    "Profit": "sum",
    "Quantity": "sum"
}).reset_index()

sales_median = product_summary["Sales"].median()
profit_median = product_summary["Profit"].median()

def product_segment(row):
    if row["Sales"] >= sales_median and row["Profit"] >= profit_median:
        return "High Sales, High Profit"
    elif row["Sales"] >= sales_median and row["Profit"] < profit_median:
        return "High Sales, Low Profit"
    elif row["Sales"] < sales_median and row["Profit"] >= profit_median:
        return "Low Sales, High Profit"
    else:
        return "Low Sales, Low Profit"

product_summary["Product Segment"] = product_summary.apply(product_segmer
```

C. Discount Impact Analysis

Create discount bins:

```
df["Discount Group"] = pd.cut(
    df["Discount"],
    bins=[-0.01, 0, 0.1, 0.2, 0.3, 0.5, 1],
    labels=["No Discount", "0-10%", "10-20%", "20-30%", "30-50%", "50%+"]
)
```

Then analyze:

```
discount_analysis = df.groupby("Discount Group")[["Sales", "Profit"]].sum

discount_analysis.plot(
    kind="bar",
    figsize=(10, 5),
    title="Sales and Profit by Discount Group"
)
```

D. Sales Forecasting Starter

Create a simple monthly sales trend and moving average.

```
monthly_sales_df = (
    df.groupby(df["Order Date"].dt.to_period("M"))["Sales"]
    .sum()
    .reset_index()
)

monthly_sales_df["Order Date"] = monthly_sales_df["Order Date"].astype(str)
monthly_sales_df["Moving Average"] = monthly_sales_df["Sales"].rolling(window=3)

monthly_sales_df.plot(
    x="Order Date",
    y=["Sales", "Moving Average"],
    kind="line",
    figsize=(12, 5),
    title="Monthly Sales with 3-Month Moving Average"
)
```

14. Suggested Notebook Structure

Use this order in your Jupyter Notebook.

1. Introduction

Explain:

- Dataset name
- Dataset source
- Project goal
- Business problem

2. Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

3. Load Dataset

```
df = pd.read_csv("data/superstore.csv")
df.head()
```

4. Data Overview

Use:

```
df.shape
df.info()
df.describe()
df.columns
```

5. Data Cleaning

Include:

- Missing values
- Duplicate records
- Date conversion
- New columns

6. KPI Summary

Show:

- Total sales
- Total profit
- Total orders
- Total customers
- Average order value
- Profit margin

7. Exploratory Data Analysis

Analyze:

- Sales trends
- Profit trends
- Category performance
- Region performance
- Customer segment performance
- Product performance
- Discount impact

8. Business Insights

Write clear findings.

Example:

- Technology generated the highest sales.
- Some sub-categories had high sales but negative profit.

- Higher discounts often reduced profit.
- Certain regions need better pricing or discount control.

9. Recommendations

Give business suggestions.

Example:

- Reduce discounts on loss-making sub-categories.
- Focus marketing on high-profit customer segments.
- Improve performance in low-profit regions.
- Promote products with both high sales and high profit.

10. Conclusion

Summarize the project in 5 to 7 lines.

15. Final Business Questions for README

Use these questions in your GitHub README:

1. What is the overall sales and profit performance of the store?
 2. Which product categories and sub-categories drive the most revenue?
 3. Which products are causing losses?
 4. Which regions and states are most profitable?
 5. Which customer segment is most valuable?
 6. How does discount affect profit?
 7. Which shipping mode is most commonly used?
 8. What are the monthly and yearly sales trends?
 9. What recommendations can improve profitability?
 10. What business actions should management take?
-

16. Example Insights You Can Aim to Find

After completing the analysis, your final insights may look like this:

1. Sales increased over time, but profit did not always increase at the same rate.
 2. Some products produced high revenue but low or negative profit.
 3. Discounts had a negative effect on profit in several categories.
 4. One region may dominate in sales, but another may have better profit margins.
 5. Certain sub-categories may need pricing review because they create losses.
 6. The company should focus on profitable products instead of only high-selling products.
 7. Customer segment analysis can help target better marketing campaigns.
-

17. Skills This Project Will Show

This project will demonstrate:

- Data loading with pandas
 - Data cleaning
 - Date handling
 - GroupBy analysis
 - Pivot tables
 - KPI calculation
 - Business problem solving
 - Data visualization
 - Trend analysis
 - Customer analysis
 - Product analysis
 - Profitability analysis
 - Insight writing
 - Recommendation writing
-

18. Final Portfolio Project Name

Use this name on GitHub or your resume:

Superstore Sales and Profitability Analysis Using Python

Resume bullet example:

Analyzed a multi-year retail sales dataset using Python, Pandas, Matplotlib, and Seaborn to identify sales trends, profit drivers, discount impact, regional performance, and loss-making products; created visualizations and business recommendations to improve profitability.

19. Minimum Plots Required for a Strong Portfolio Project

If you do not want to create all plots, create at least these 10:

1. Monthly Sales Trend
 2. Yearly Sales and Profit
 3. Sales by Category
 4. Profit by Category
 5. Sales by Sub-Category
 6. Profit by Sub-Category
 7. Sales by Region
 8. Profit by Region
 9. Discount vs Profit
 10. Top 10 Loss-Making Products
-

20. Final Project Difficulty

Level: Beginner to Intermediate

Estimated time: 2 to 4 days

Best for: Data analyst portfolio

Main tool: Pandas

Visualization tools: Matplotlib and Seaborn

Dataset language: English

21. Project Evaluation Checklist

Before publishing the project, make sure you have:

- Clean notebook structure

- Clear markdown explanations
 - No unnecessary code output
 - At least 10 useful plots
 - At least 8 business questions answered
 - KPI summary section
 - Final insights section
 - Final recommendations section
 - README file
 - Dataset source mentioned
 - Clean GitHub repository
-

22. Suggested GitHub Repository Description

Repository name: `superstore-sales-analysis-pandas`

Description:

A data analysis project using Python Pandas to explore sales, profit, customer behavior, product performance, discount impact, and regional trends in a Superstore retail dataset.

23. Suggested README Summary

This project analyzes a Superstore retail dataset using Python, Pandas, Matplotlib, and Seaborn. The main objective is to understand sales performance, profitability, product trends, regional behavior, customer segments, and discount impact. The project includes data cleaning, feature engineering, exploratory data analysis, KPI calculation, visualizations, business insights, and recommendations.

24. Suggested Requirements File

```
pandas
numpy
matplotlib
seaborn
jupyter
```

